

Alpha Student

Original Blue Heading

This paragraph must remain in Calibri 14 with blue text.

Metric	Value
Preserved table	42

1

Inflation in the United States and Its Effects on Local People

Name

University

subject code and name

Professor _____

Due date

Beta Student

Second Source Title

A highlighted sentence from the second source.

IGNORE

Global Warming: Causes, Consequences, and Global Responses

Global warming refers to the long-term increase in Earth's average surface temperature, primarily resulting from human activities that increase concentrations of heat-trapping greenhouse gases in the atmosphere. Although Earth's climate has always experienced natural variations, the warming observed since the Industrial Revolution differs substantially in its speed, scale, and principal causes. Scientific observations from surface monitoring stations, satellites, ocean measurements, glaciers, ice cores, and other sources provide strong evidence that the planet is warming and that human activities are the dominant cause. The Intergovernmental Panel on Climate Change (IPCC, 2023) concluded that human activities, principally greenhouse gas emissions, have unequivocally caused global warming. Average global surface temperature during 2011–2020 was approximately 1.1°C above the 1850–1900 average. This seemingly modest change has already produced significant effects on weather systems, ecosystems, human health, agriculture, water availability, and economies. Global warming is therefore not simply an environmental concern; it is also a social, economic, developmental, and public-health challenge requiring coordinated action at local, national, and international levels.

Causes and Scientific Evidence of Global Warming

The principal cause of modern global warming is the enhanced greenhouse effect created by increasing concentrations of carbon dioxide, methane, nitrous oxide, and other greenhouse gases.

The natural greenhouse effect is essential for life because gases in the atmosphere retain part of the heat that would otherwise escape into space. Human activity, however, has strengthened this process by releasing additional greenhouse gases. Fossil-fuel combustion for electricity, transportation, heating, industrial activity, and manufacturing remains a major source of carbon dioxide emissions. Agriculture, livestock production, waste management, fossil-fuel extraction, and other activities contribute substantial methane emissions, while agricultural fertilizers and certain industrial processes release nitrous oxide. Deforestation further contributes to the problem because removing forests releases stored carbon and reduces the ability of ecosystems to absorb carbon dioxide from the atmosphere.

The rapid expansion of industrial production and energy consumption since the nineteenth century has therefore altered the composition of the atmosphere. NASA explains that the current warming trend is clearly linked to human activities since the mid-1800s and is occurring at a rate not observed over many recent millennia (National Aeronautics and Space Administration [NASA], n.d.-a). Atmospheric carbon dioxide concentrations can be reconstructed through ice cores and compared with modern direct measurements, providing scientists with a long-term record of environmental changes. These records demonstrate that recent increases in atmospheric carbon dioxide are substantially different from the natural fluctuations associated with previous climate cycles.

Natural factors such as volcanic eruptions, changes in solar activity, and variations in Earth's orbit can influence the climate. However, these factors cannot explain the magnitude and pattern of warming observed during the modern industrial era. The IPCC (2023) estimated that the best

estimate of human-caused warming between 1850–1900 and 2010–2019 was approximately 1.07°C. In comparison, natural solar and volcanic drivers produced only very small changes in global temperature over the same broad period. The evidence therefore contradicts claims that contemporary warming can primarily be explained by natural climate cycles.

Multiple independent observations confirm the warming trend. Global surface-temperature datasets developed by different scientific organizations use somewhat different methodologies but consistently demonstrate substantial long-term warming. NASA reports that recent decades have become progressively warmer and that the temperature increase since 1970 has occurred faster than during any other 50-year period in at least the last 2,000 years (NASA, n.d.-b). Scientists also examine indirect indicators such as tree rings, coral formations, ice cores, sediments, and glacier records. Agreement among these independent sources significantly strengthens confidence in the scientific conclusion that the planet is warming.

Changes in the oceans and cryosphere provide additional evidence. Oceans absorb most of the additional heat accumulated in the climate system. According to the IPCC (2023), ocean warming accounted for approximately 91% of the heating within the climate system, while smaller proportions were associated with warming land, melting ice, and atmospheric warming. Glaciers and ice sheets have lost mass, Arctic sea ice has declined, and global sea levels have risen as a consequence of both melting land ice and the thermal expansion of seawater. Human influence has also contributed to changes in extreme weather conditions. Hot extremes and heat waves have become more frequent and intense across most land regions since the 1950s, while marine heat waves have approximately doubled in frequency since the 1980s (IPCC, 2023).

Global warming is consequently better understood as a disruption of the entire climate system rather than simply an increase in average temperature. Changes in atmospheric and ocean temperatures affect precipitation patterns, drought conditions, storms, ecosystems, ice, and sea levels. Because these components interact, warming can produce complex and cascading consequences. The scientific evidence gathered through independent measurements and decades of research demonstrates that limiting further warming requires substantial reductions in human-generated greenhouse gas emissions.

Environmental, Social, and Economic Impacts and Global Responses

The effects of global warming are already visible throughout natural and human systems. One of the clearest consequences is the increasing frequency and severity of extreme heat. Higher average temperatures make extreme heat events more probable and can increase their intensity and duration. Heat waves create serious risks for older adults, children, outdoor workers, and people with existing illnesses. Extreme heat can cause dehydration, heat exhaustion, heat stroke, cardiovascular stress, and death. The World Health Organization (WHO, 2023) identifies climate change as a major threat to human health because it affects clean air, safe water, food supplies, shelter, livelihoods, and healthcare systems. Climate change is also associated with health risks arising from floods, storms, wildfires, droughts, infectious diseases, and mental-health pressures.

Agriculture and food security are particularly vulnerable because crop productivity depends on temperature, water availability, soil quality, predictable seasonal conditions, and protection from

extreme weather. Excessive heat can reduce crop yields, while drought can create water shortages and damage livestock production. Conversely, intense rainfall and flooding can destroy crops, erode soil, damage transportation networks, and disrupt food distribution. Changes in climatic conditions may also allow agricultural pests and plant diseases to expand into new areas. These effects can increase food prices and intensify malnutrition, especially in low-income regions where households already spend a significant proportion of their income on food.

Water resources are similarly affected. Changing rainfall patterns may increase drought risks in some regions while increasing flooding in others. Mountain glaciers, which provide seasonal freshwater to many communities, are retreating as temperatures rise. Coastal freshwater supplies may also face saltwater intrusion as sea levels increase. Water scarcity can affect agriculture, electricity generation, sanitation, industry, and household consumption, potentially increasing competition over already limited resources.

Ecosystems are being transformed as species respond to changing temperatures, precipitation patterns, wildfire conditions, and habitat availability. Some species can migrate toward cooler regions or higher elevations, whereas others cannot adapt or move rapidly enough. Changes in species distributions can alter food webs and ecological relationships. Abrahms et al. (2023) found that climatic changes can also intensify human-wildlife conflicts by altering animal behavior, resource availability, and patterns of interaction between wildlife and human populations. These ecological disruptions matter because healthy ecosystems provide important services, including pollination, water regulation, soil protection, carbon storage, fisheries, and cultural benefits.

Coastal regions face another major threat from rising sea levels. Sea-level rise increases the likelihood of coastal flooding, erosion, saltwater intrusion, and damage from storm surges. Small island states, low-lying coastal communities, and densely populated river deltas face particularly serious risks. Global warming may therefore increase displacement and migration when communities lose homes, agricultural land, freshwater supplies, or livelihoods. Importantly, climate impacts are not distributed equally. The IPCC (2023) emphasizes that communities that have historically contributed the least to climate change are often disproportionately affected. Poverty, inadequate infrastructure, limited healthcare access, weak institutions, and geographical exposure can make communities significantly more vulnerable. Climate change therefore raises important questions about environmental justice and international responsibility.

Addressing global warming requires both mitigation and adaptation. Mitigation focuses on reducing the causes of climate change, whereas adaptation involves preparing societies and ecosystems for impacts that can no longer be completely avoided. Effective mitigation requires reducing dependence on fossil fuels, expanding renewable and low-carbon energy, improving energy efficiency, electrifying appropriate forms of transportation and heating, reducing methane emissions, protecting forests, restoring degraded ecosystems, and improving agricultural and industrial practices. Technological innovation can support this transition through improved battery storage, electricity grids, renewable-energy systems, energy-efficient buildings, low-carbon industrial processes, and carbon-management technologies.

Governments also play a central role through regulations, public investment, infrastructure development, research funding, energy policies, and international cooperation. The United Nations Environment Programme (UNEP, 2025) continues to assess the gap between countries' emission commitments and the reductions required to keep global warming consistent with internationally agreed temperature goals. Closing this emissions gap requires stronger implementation rather than relying only on distant targets. Businesses also have responsibilities because corporate investment decisions influence energy systems, supply chains, transportation networks, manufacturing technologies, and patterns of resource consumption.

Adaptation is equally important because a certain degree of additional climate change is already unavoidable due to past and continuing emissions. Adaptation measures include heat-action plans, improved flood protection, stronger building standards, drought-resistant crops, efficient irrigation systems, wildfire preparedness, climate-resilient healthcare facilities, early-warning systems, urban green spaces, and improved disaster-management capabilities. Adaptation planning should prioritize populations facing the greatest risks and having the fewest resources to respond.

Individual behavior can contribute through energy conservation, lower-emission transportation, reduction of food and material waste, and sustainable consumption. Nevertheless, global warming cannot be solved solely through individual lifestyle choices. Large-scale transformation of energy, transportation, industry, agriculture, land use, and infrastructure is necessary. Governments, businesses, communities, researchers, and individuals therefore have complementary responsibilities.

Ultimately, global warming represents one of the most significant long-term challenges facing humanity. The scientific evidence demonstrates clearly that human-generated greenhouse gas emissions are the dominant cause of recent warming, while observed changes in temperature, oceans, ice, weather extremes, ecosystems, and sea levels show that its consequences are already occurring. Continued warming will increase risks to health, biodiversity, food and water security, economic development, infrastructure, and vulnerable communities. However, the severity of future impacts is not predetermined. Rapid reductions in greenhouse gas emissions, combined with effective adaptation, technological innovation, ecosystem protection, and international cooperation, can substantially reduce future damages. Global warming therefore demands immediate and sustained action based on scientific evidence, social responsibility, and recognition that decisions made today will influence environmental and human conditions for generations to come.

Global Warming: Causes, Consequences, and Global Responses

Global warming refers to the long-term increase in Earth's average surface temperature, primarily resulting from human activities that increase concentrations of heat-trapping greenhouse gases in the atmosphere. Although Earth's climate has always experienced natural variations, the warming observed since the Industrial Revolution differs substantially in its speed, scale, and principal causes. Scientific observations from surface monitoring stations, satellites, ocean measurements, glaciers, ice cores, and other sources provide strong evidence that the planet is warming and that human activities are the dominant cause. The

Intergovernmental Panel on Climate Change (IPCC, 2023) concluded that human activities, principally greenhouse gas emissions, have unequivocally caused global warming. Average global surface temperature during 2011–2020 was approximately 1.1°C above the 1850–1900 average. This seemingly modest change has already produced significant effects on weather systems, ecosystems, human health, agriculture, water availability, and economies. Global warming is therefore not simply an environmental concern; it is also a social, economic, developmental, and public-health challenge requiring coordinated action at local, national, and international levels.

Causes and Scientific Evidence of Global Warming

The principal cause of modern global warming is the enhanced greenhouse effect created by increasing concentrations of carbon dioxide, methane, nitrous oxide, and other greenhouse gases. The natural greenhouse effect is essential for life because gases in the atmosphere retain part of the heat that would otherwise escape into space. Human activity, however, has strengthened this process by releasing additional greenhouse gases. Fossil-fuel combustion for electricity, transportation, heating, industrial activity, and manufacturing remains a major source of carbon dioxide emissions. Agriculture, livestock production, waste management, fossil-fuel extraction, and other activities contribute substantial methane emissions, while agricultural fertilizers and certain industrial processes release nitrous oxide. Deforestation further contributes to the problem because removing forests releases stored carbon and reduces the ability of ecosystems to absorb carbon dioxide from the atmosphere.

The rapid expansion of industrial production and energy consumption since the nineteenth century has therefore altered the composition of the atmosphere. NASA explains that the current warming trend is clearly linked to human activities since the mid-1800s and is occurring at a rate not observed over many recent millennia (National Aeronautics and Space Administration [NASA], n.d.-a). Atmospheric carbon dioxide concentrations can be reconstructed through ice cores and compared with modern direct measurements, providing scientists with a long-term record of environmental changes. These records demonstrate that recent increases in atmospheric carbon dioxide are substantially different from the natural fluctuations associated with previous climate cycles.

Natural factors such as volcanic eruptions, changes in solar activity, and variations in Earth's orbit can influence the climate. However, these factors cannot explain the magnitude and pattern of warming observed during the modern industrial era. The IPCC (2023) estimated that the best estimate of human-caused warming between 1850–1900 and 2010–2019 was approximately 1.07°C. In comparison, natural solar and volcanic drivers produced only very small changes in global temperature over the same broad period. The evidence therefore contradicts claims that contemporary warming can primarily be explained by natural climate cycles.

Multiple independent observations confirm the warming trend. Global surface-temperature datasets developed by different scientific organizations use somewhat different methodologies but consistently demonstrate substantial long-term warming. NASA reports that recent decades have become progressively warmer and that the temperature increase since 1970 has occurred faster than during any other 50-year period in at least the last 2,000 years (NASA, n.d.-b).

Scientists also examine indirect indicators such as tree rings, coral formations, ice cores, sediments, and glacier records. Agreement among these independent sources significantly strengthens confidence in the scientific conclusion that the planet is warming.

Changes in the oceans and cryosphere provide additional evidence. Oceans absorb most of the additional heat accumulated in the climate system. According to the IPCC (2023), ocean warming accounted for approximately 91% of the heating within the climate system, while smaller proportions were associated with warming land, melting ice, and atmospheric warming. Glaciers and ice sheets have lost mass, Arctic sea ice has declined, and global sea levels have risen as a consequence of both melting land ice and the thermal expansion of seawater. Human influence has also contributed to changes in extreme weather conditions. Hot extremes and heat waves have become more frequent and intense across most land regions since the 1950s, while marine heat waves have approximately doubled in frequency since the 1980s (IPCC, 2023).

Global warming is consequently better understood as a disruption of the entire climate system rather than simply an increase in average temperature. Changes in atmospheric and ocean temperatures affect precipitation patterns, drought conditions, storms, ecosystems, ice, and sea levels. Because these components interact, warming can produce complex and cascading consequences. The scientific evidence gathered through independent measurements and decades of research demonstrates that limiting further warming requires substantial reductions in human-generated greenhouse gas emissions.

Environmental, Social, and Economic Impacts and Global Responses

The effects of global warming are already visible throughout natural and human systems. One of the clearest consequences is the increasing frequency and severity of extreme heat. Higher average temperatures make extreme heat events more probable and can increase their intensity and duration. Heat waves create serious risks for older adults, children, outdoor workers, and people with existing illnesses. Extreme heat can cause dehydration, heat exhaustion, heat stroke, cardiovascular stress, and death. The World Health Organization (WHO, 2023) identifies climate change as a major threat to human health because it affects clean air, safe water, food supplies, shelter, livelihoods, and healthcare systems. Climate change is also associated with health risks arising from floods, storms, wildfires, droughts, infectious diseases, and mental-health pressures.

Agriculture and food security are particularly vulnerable because crop productivity depends on temperature, water availability, soil quality, predictable seasonal conditions, and protection from extreme weather. Excessive heat can reduce crop yields, while drought can create water shortages and damage livestock production. Conversely, intense rainfall and flooding can destroy crops, erode soil, damage transportation networks, and disrupt food distribution. Changes in climatic conditions may also allow agricultural pests and plant diseases to expand into new areas. These effects can increase food prices and intensify malnutrition, especially in low-income regions where households already spend a significant proportion of their income on food.

Water resources are similarly affected. Changing rainfall patterns may increase drought risks in some regions while increasing flooding in others. Mountain glaciers, which provide seasonal freshwater to many communities, are retreating as temperatures rise. Coastal freshwater supplies

may also face saltwater intrusion as sea levels increase. Water scarcity can affect agriculture, electricity generation, sanitation, industry, and household consumption, potentially increasing competition over already limited resources.

Ecosystems are being transformed as species respond to changing temperatures, precipitation patterns, wildfire conditions, and habitat availability. Some species can migrate toward cooler regions or higher elevations, whereas others cannot adapt or move rapidly enough. Changes in species distributions can alter food webs and ecological relationships. Abrahms et al. (2023) found that climatic changes can also intensify human-wildlife conflicts by altering animal behavior, resource availability, and patterns of interaction between wildlife and human populations. These ecological disruptions matter because healthy ecosystems provide important services, including pollination, water regulation, soil protection, carbon storage, fisheries, and cultural benefits.

Coastal regions face another major threat from rising sea levels. Sea-level rise increases the likelihood of coastal flooding, erosion, saltwater intrusion, and damage from storm surges. Small island states, low-lying coastal communities, and densely populated river deltas face particularly serious risks. Global warming may therefore increase displacement and migration when communities lose homes, agricultural land, freshwater supplies, or livelihoods. Importantly, climate impacts are not distributed equally. The IPCC (2023) emphasizes that communities that have historically contributed the least to climate change are often disproportionately affected. Poverty, inadequate infrastructure, limited healthcare access, weak institutions, and geographical

exposure can make communities significantly more vulnerable. Climate change therefore raises important questions about environmental justice and international responsibility.

Addressing global warming requires both mitigation and adaptation. Mitigation focuses on reducing the causes of climate change, whereas adaptation involves preparing societies and ecosystems for impacts that can no longer be completely avoided. Effective mitigation requires reducing dependence on fossil fuels, expanding renewable and low-carbon energy, improving energy efficiency, electrifying appropriate forms of transportation and heating, reducing methane emissions, protecting forests, restoring degraded ecosystems, and improving agricultural and industrial practices. Technological innovation can support this transition through improved battery storage, electricity grids, renewable-energy systems, energy-efficient buildings, low-carbon industrial processes, and carbon-management technologies.

Governments also play a central role through regulations, public investment, infrastructure development, research funding, energy policies, and international cooperation. The United Nations Environment Programme (UNEP, 2025) continues to assess the gap between countries' emission commitments and the reductions required to keep global warming consistent with internationally agreed temperature goals. Closing this emissions gap requires stronger implementation rather than relying only on distant targets. Businesses also have responsibilities because corporate investment decisions influence energy systems, supply chains, transportation networks, manufacturing technologies, and patterns of resource consumption.

Adaptation is equally important because a certain degree of additional climate change is already unavoidable due to past and continuing emissions. Adaptation measures include heat-action plans, improved flood protection, stronger building standards, drought-resistant crops, efficient irrigation systems, wildfire preparedness, climate-resilient healthcare facilities, early-warning systems, urban green spaces, and improved disaster-management capabilities. Adaptation planning should prioritize populations facing the greatest risks and having the fewest resources to respond.

Individual behavior can contribute through energy conservation, lower-emission transportation, reduction of food and material waste, and sustainable consumption. Nevertheless, global warming cannot be solved solely through individual lifestyle choices. Large-scale transformation of energy, transportation, industry, agriculture, land use, and infrastructure is necessary. Governments, businesses, communities, researchers, and individuals therefore have complementary responsibilities.

Ultimately, global warming represents one of the most significant long-term challenges facing humanity. The scientific evidence demonstrates clearly that human-generated greenhouse gas emissions are the dominant cause of recent warming, while observed changes in temperature, oceans, ice, weather extremes, ecosystems, and sea levels show that its consequences are already occurring. Continued warming will increase risks to health, biodiversity, food and water security, economic development, infrastructure, and vulnerable communities. However, the severity of future impacts is not predetermined. Rapid reductions in greenhouse gas emissions, combined with effective adaptation, technological innovation, ecosystem protection, and international

cooperation, can substantially reduce future damages. Global warming therefore demands immediate and sustained action based on scientific evidence, social responsibility, and recognition that decisions made today will influence environmental and human conditions for generations to come.

Global Warming: Causes, Consequences, and Global Responses

Global warming refers to the long-term increase in Earth's average surface temperature, primarily resulting from human activities that increase concentrations of heat-trapping greenhouse gases in the atmosphere. Although Earth's climate has always experienced natural variations, the warming observed since the Industrial Revolution differs substantially in its speed, scale, and principal causes. Scientific observations from surface monitoring stations, satellites, ocean measurements, glaciers, ice cores, and other sources provide strong evidence that the planet is warming and that human activities are the dominant cause. The Intergovernmental Panel on Climate Change (IPCC, 2023) concluded that human activities, principally greenhouse gas emissions, have unequivocally caused global warming. Average global surface temperature during 2011–2020 was approximately 1.1°C above the 1850–1900 average. This seemingly modest change has already produced significant effects on weather systems, ecosystems, human health, agriculture, water availability, and economies. Global warming is therefore not simply an environmental concern; it is also a social, economic, developmental, and public-health challenge requiring coordinated action at local, national, and international levels.

Causes and Scientific Evidence of Global Warming

The principal cause of modern global warming is the enhanced greenhouse effect created by increasing concentrations of carbon dioxide, methane, nitrous oxide, and other greenhouse gases. The natural greenhouse effect is essential for life because gases in the atmosphere retain part of the heat that would otherwise escape into space. Human activity, however, has strengthened this process by releasing additional greenhouse gases. Fossil-fuel combustion for electricity, transportation, heating, industrial activity, and manufacturing remains a major source of carbon dioxide emissions. Agriculture, livestock production, waste management, fossil-fuel extraction, and other activities contribute substantial methane emissions, while agricultural fertilizers and certain industrial processes release nitrous oxide. Deforestation further contributes to the problem because removing forests releases stored carbon and reduces the ability of ecosystems to absorb carbon dioxide from the atmosphere.

The rapid expansion of industrial production and energy consumption since the nineteenth century has therefore altered the composition of the atmosphere. NASA explains that the current warming trend is clearly linked to human activities since the mid-1800s and is occurring at a rate not observed over many recent millennia (National Aeronautics and Space Administration [NASA], n.d.-a). Atmospheric carbon dioxide concentrations can be reconstructed through ice cores and compared with modern direct measurements, providing scientists with a long-term record of environmental changes. These records demonstrate that recent increases in atmospheric carbon dioxide are substantially different from the natural fluctuations associated with previous climate cycles.

Natural factors such as volcanic eruptions, changes in solar activity, and variations in Earth's orbit can influence the climate. However, these factors cannot explain the magnitude and pattern of warming observed during the modern industrial era. The IPCC (2023) estimated that the best estimate of human-caused warming between 1850–1900 and 2010–2019 was approximately 1.07°C. In comparison, natural solar and volcanic drivers produced only very small changes in global temperature over the same broad period. The evidence therefore contradicts claims that contemporary warming can primarily be explained by natural climate cycles.

Multiple independent observations confirm the warming trend. Global surface-temperature datasets developed by different scientific organizations use somewhat different methodologies but consistently demonstrate substantial long-term warming. NASA reports that recent decades have become progressively warmer and that the temperature increase since 1970 has occurred faster than during any other 50-year period in at least the last 2,000 years (NASA, n.d.-b). Scientists also examine indirect indicators such as tree rings, coral formations, ice cores, sediments, and glacier records. Agreement among these independent sources significantly strengthens confidence in the scientific conclusion that the planet is warming.

Changes in the oceans and cryosphere provide additional evidence. Oceans absorb most of the additional heat accumulated in the climate system. According to the IPCC (2023), ocean warming accounted for approximately 91% of the heating within the climate system, while smaller proportions were associated with warming land, melting ice, and atmospheric warming. Glaciers and ice sheets have lost mass, Arctic sea ice has declined, and global sea levels have

risen as a consequence of both melting land ice and the thermal expansion of seawater. Human influence has also contributed to changes in extreme weather conditions. Hot extremes and heat waves have become more frequent and intense across most land regions since the 1950s, while marine heat waves have approximately doubled in frequency since the 1980s (IPCC, 2023).

Global warming is consequently better understood as a disruption of the entire climate system rather than simply an increase in average temperature. Changes in atmospheric and ocean temperatures affect precipitation patterns, drought conditions, storms, ecosystems, ice, and sea levels. Because these components interact, warming can produce complex and cascading consequences. The scientific evidence gathered through independent measurements and decades of research demonstrates that limiting further warming requires substantial reductions in human-generated greenhouse gas emissions.

Environmental, Social, and Economic Impacts and Global Responses

The effects of global warming are already visible throughout natural and human systems. One of the clearest consequences is the increasing frequency and severity of extreme heat. Higher average temperatures make extreme heat events more probable and can increase their intensity and duration. Heat waves create serious risks for older adults, children, outdoor workers, and people with existing illnesses. Extreme heat can cause dehydration, heat exhaustion, heat stroke, cardiovascular stress, and death. The World Health Organization (WHO, 2023) identifies climate change as a major threat to human health because it affects clean air, safe water, food supplies,

shelter, livelihoods, and healthcare systems. Climate change is also associated with health risks arising from floods, storms, wildfires, droughts, infectious diseases, and mental-health pressures.

Agriculture and food security are particularly vulnerable because crop productivity depends on temperature, water availability, soil quality, predictable seasonal conditions, and protection from extreme weather. Excessive heat can reduce crop yields, while drought can create water shortages and damage livestock production. Conversely, intense rainfall and flooding can destroy crops, erode soil, damage transportation networks, and disrupt food distribution. Changes in climatic conditions may also allow agricultural pests and plant diseases to expand into new areas. These effects can increase food prices and intensify malnutrition, especially in low-income regions where households already spend a significant proportion of their income on food.

Water resources are similarly affected. Changing rainfall patterns may increase drought risks in some regions while increasing flooding in others. Mountain glaciers, which provide seasonal freshwater to many communities, are retreating as temperatures rise. Coastal freshwater supplies may also face saltwater intrusion as sea levels increase. Water scarcity can affect agriculture, electricity generation, sanitation, industry, and household consumption, potentially increasing competition over already limited resources.

Ecosystems are being transformed as species respond to changing temperatures, precipitation patterns, wildfire conditions, and habitat availability. Some species can migrate toward cooler regions or higher elevations, whereas others cannot adapt or move rapidly enough. Changes in species distributions can alter food webs and ecological relationships. Abrahms et al. (2023)

found that climatic changes can also intensify human-wildlife conflicts by altering animal behavior, resource availability, and patterns of interaction between wildlife and human populations. These ecological disruptions matter because healthy ecosystems provide important services, including pollination, water regulation, soil protection, carbon storage, fisheries, and cultural benefits.

Coastal regions face another major threat from rising sea levels. Sea-level rise increases the likelihood of coastal flooding, erosion, saltwater intrusion, and damage from storm surges. Small island states, low-lying coastal communities, and densely populated river deltas face particularly serious risks. Global warming may therefore increase displacement and migration when communities lose homes, agricultural land, freshwater supplies, or livelihoods. Importantly, climate impacts are not distributed equally. The IPCC (2023) emphasizes that communities that have historically contributed the least to climate change are often disproportionately affected. Poverty, inadequate infrastructure, limited healthcare access, weak institutions, and geographical exposure can make communities significantly more vulnerable. Climate change therefore raises important questions about environmental justice and international responsibility.

Addressing global warming requires both mitigation and adaptation. Mitigation focuses on reducing the causes of climate change, whereas adaptation involves preparing societies and ecosystems for impacts that can no longer be completely avoided. Effective mitigation requires reducing dependence on fossil fuels, expanding renewable and low-carbon energy, improving energy efficiency, electrifying appropriate forms of transportation and heating, reducing methane emissions, protecting forests, restoring degraded ecosystems, and improving agricultural and

industrial practices. Technological innovation can support this transition through improved battery storage, electricity grids, renewable-energy systems, energy-efficient buildings, low-carbon industrial processes, and carbon-management technologies.

Governments also play a central role through regulations, public investment, infrastructure development, research funding, energy policies, and international cooperation. The United Nations Environment Programme (UNEP, 2025) continues to assess the gap between countries' emission commitments and the reductions required to keep global warming consistent with internationally agreed temperature goals. Closing this emissions gap requires stronger implementation rather than relying only on distant targets. Businesses also have responsibilities because corporate investment decisions influence energy systems, supply chains, transportation networks, manufacturing technologies, and patterns of resource consumption.

Adaptation is equally important because a certain degree of additional climate change is already unavoidable due to past and continuing emissions. Adaptation measures include heat-action plans, improved flood protection, stronger building standards, drought-resistant crops, efficient irrigation systems, wildfire preparedness, climate-resilient healthcare facilities, early-warning systems, urban green spaces, and improved disaster-management capabilities. Adaptation planning should prioritize populations facing the greatest risks and having the fewest resources to respond.

Individual behavior can contribute through energy conservation, lower-emission transportation, reduction of food and material waste, and sustainable consumption. Nevertheless, global

warming cannot be solved solely through individual lifestyle choices. Large-scale transformation of energy, transportation, industry, agriculture, land use, and infrastructure is necessary. Governments, businesses, communities, researchers, and individuals therefore have complementary responsibilities.

Ultimately, global warming represents one of the most significant long-term challenges facing humanity. The scientific evidence demonstrates clearly that human-generated greenhouse gas emissions are the dominant cause of recent warming, while observed changes in temperature, oceans, ice, weather extremes, ecosystems, and sea levels show that its consequences are already occurring. Continued warming will increase risks to health, biodiversity, food and water security, economic development, infrastructure, and vulnerable communities. However, the severity of future impacts is not predetermined. Rapid reductions in greenhouse gas emissions, combined with effective adaptation, technological innovation, ecosystem protection, and international cooperation, can substantially reduce future damages. Global warming therefore demands immediate and sustained action based on scientific evidence, social responsibility, and recognition that decisions made today will influence environmental and human conditions for generations to come.

Global Warming: Causes, Consequences, and Global Responses

Global warming refers to the long-term increase in Earth's average surface temperature, primarily resulting from human activities that increase concentrations of heat-trapping greenhouse gases in the atmosphere. Although Earth's climate has always experienced natural

variations, the warming observed since the Industrial Revolution differs substantially in its speed, scale, and principal causes. Scientific observations from surface monitoring stations, satellites, ocean measurements, glaciers, ice cores, and other sources provide strong evidence that the planet is warming and that human activities are the dominant cause. The Intergovernmental Panel on Climate Change (IPCC, 2023) concluded that human activities, principally greenhouse gas emissions, have unequivocally caused global warming. Average global surface temperature during 2011–2020 was approximately 1.1°C above the 1850–1900 average. This seemingly modest change has already produced significant effects on weather systems, ecosystems, human health, agriculture, water availability, and economies. Global warming is therefore not simply an environmental concern; it is also a social, economic, developmental, and public-health challenge requiring coordinated action at local, national, and international levels.

Causes and Scientific Evidence of Global Warming

The principal cause of modern global warming is the enhanced greenhouse effect created by increasing concentrations of carbon dioxide, methane, nitrous oxide, and other greenhouse gases. The natural greenhouse effect is essential for life because gases in the atmosphere retain part of the heat that would otherwise escape into space. Human activity, however, has strengthened this process by releasing additional greenhouse gases. Fossil-fuel combustion for electricity, transportation, heating, industrial activity, and manufacturing remains a major source of carbon dioxide emissions. Agriculture, livestock production, waste management, fossil-fuel extraction, and other activities contribute substantial methane emissions, while agricultural fertilizers and

certain industrial processes release nitrous oxide. Deforestation further contributes to the problem because removing forests releases stored carbon and reduces the ability of ecosystems to absorb carbon dioxide from the atmosphere.

The rapid expansion of industrial production and energy consumption since the nineteenth century has therefore altered the composition of the atmosphere. NASA explains that the current warming trend is clearly linked to human activities since the mid-1800s and is occurring at a rate not observed over many recent millennia (National Aeronautics and Space Administration [NASA], n.d.-a). Atmospheric carbon dioxide concentrations can be reconstructed through ice cores and compared with modern direct measurements, providing scientists with a long-term record of environmental changes. These records demonstrate that recent increases in atmospheric carbon dioxide are substantially different from the natural fluctuations associated with previous climate cycles.

Natural factors such as volcanic eruptions, changes in solar activity, and variations in Earth's orbit can influence the climate. However, these factors cannot explain the magnitude and pattern of warming observed during the modern industrial era. The IPCC (2023) estimated that the best estimate of human-caused warming between 1850–1900 and 2010–2019 was approximately 1.07°C. In comparison, natural solar and volcanic drivers produced only very small changes in global temperature over the same broad period. The evidence therefore contradicts claims that contemporary warming can primarily be explained by natural climate cycles.

Multiple independent observations confirm the warming trend. Global surface-temperature datasets developed by different scientific organizations use somewhat different methodologies but consistently demonstrate substantial long-term warming. NASA reports that recent decades have become progressively warmer and that the temperature increase since 1970 has occurred faster than during any other 50-year period in at least the last 2,000 years (NASA, n.d.-b). Scientists also examine indirect indicators such as tree rings, coral formations, ice cores, sediments, and glacier records. Agreement among these independent sources significantly strengthens confidence in the scientific conclusion that the planet is warming.

Changes in the oceans and cryosphere provide additional evidence. Oceans absorb most of the additional heat accumulated in the climate system. According to the IPCC (2023), ocean warming accounted for approximately 91% of the heating within the climate system, while smaller proportions were associated with warming land, melting ice, and atmospheric warming. Glaciers and ice sheets have lost mass, Arctic sea ice has declined, and global sea levels have risen as a consequence of both melting land ice and the thermal expansion of seawater. Human influence has also contributed to changes in extreme weather conditions. Hot extremes and heat waves have become more frequent and intense across most land regions since the 1950s, while marine heat waves have approximately doubled in frequency since the 1980s (IPCC, 2023).

Global warming is consequently better understood as a disruption of the entire climate system rather than simply an increase in average temperature. Changes in atmospheric and ocean temperatures affect precipitation patterns, drought conditions, storms, ecosystems, ice, and sea levels. Because these components interact, warming can produce complex and cascading

consequences. The scientific evidence gathered through independent measurements and decades of research demonstrates that limiting further warming requires substantial reductions in human-generated greenhouse gas emissions.

Environmental, Social, and Economic Impacts and Global Responses

The effects of global warming are already visible throughout natural and human systems. One of the clearest consequences is the increasing frequency and severity of extreme heat. Higher average temperatures make extreme heat events more probable and can increase their intensity and duration. Heat waves create serious risks for older adults, children, outdoor workers, and people with existing illnesses. Extreme heat can cause dehydration, heat exhaustion, heat stroke, cardiovascular stress, and death. The World Health Organization (WHO, 2023) identifies climate change as a major threat to human health because it affects clean air, safe water, food supplies, shelter, livelihoods, and healthcare systems. Climate change is also associated with health risks arising from floods, storms, wildfires, droughts, infectious diseases, and mental-health pressures.

Agriculture and food security are particularly vulnerable because crop productivity depends on temperature, water availability, soil quality, predictable seasonal conditions, and protection from extreme weather. Excessive heat can reduce crop yields, while drought can create water shortages and damage livestock production. Conversely, intense rainfall and flooding can destroy crops, erode soil, damage transportation networks, and disrupt food distribution. Changes in climatic conditions may also allow agricultural pests and plant diseases to expand into new areas.

These effects can increase food prices and intensify malnutrition, especially in low-income regions where households already spend a significant proportion of their income on food.

Water resources are similarly affected. Changing rainfall patterns may increase drought risks in some regions while increasing flooding in others. Mountain glaciers, which provide seasonal freshwater to many communities, are retreating as temperatures rise. Coastal freshwater supplies may also face saltwater intrusion as sea levels increase. Water scarcity can affect agriculture, electricity generation, sanitation, industry, and household consumption, potentially increasing competition over already limited resources.

Ecosystems are being transformed as species respond to changing temperatures, precipitation patterns, wildfire conditions, and habitat availability. Some species can migrate toward cooler regions or higher elevations, whereas others cannot adapt or move rapidly enough. Changes in species distributions can alter food webs and ecological relationships. Abrahms et al. (2023) found that climatic changes can also intensify human-wildlife conflicts by altering animal behavior, resource availability, and patterns of interaction between wildlife and human populations. These ecological disruptions matter because healthy ecosystems provide important services, including pollination, water regulation, soil protection, carbon storage, fisheries, and cultural benefits.

Coastal regions face another major threat from rising sea levels. Sea-level rise increases the likelihood of coastal flooding, erosion, saltwater intrusion, and damage from storm surges. Small island states, low-lying coastal communities, and densely populated river deltas face particularly

serious risks. Global warming may therefore increase displacement and migration when communities lose homes, agricultural land, freshwater supplies, or livelihoods. Importantly, climate impacts are not distributed equally. The IPCC (2023) emphasizes that communities that have historically contributed the least to climate change are often disproportionately affected. Poverty, inadequate infrastructure, limited healthcare access, weak institutions, and geographical exposure can make communities significantly more vulnerable. Climate change therefore raises important questions about environmental justice and international responsibility.

Addressing global warming requires both mitigation and adaptation. Mitigation focuses on reducing the causes of climate change, whereas adaptation involves preparing societies and ecosystems for impacts that can no longer be completely avoided. Effective mitigation requires reducing dependence on fossil fuels, expanding renewable and low-carbon energy, improving energy efficiency, electrifying appropriate forms of transportation and heating, reducing methane emissions, protecting forests, restoring degraded ecosystems, and improving agricultural and industrial practices. Technological innovation can support this transition through improved battery storage, electricity grids, renewable-energy systems, energy-efficient buildings, low-carbon industrial processes, and carbon-management technologies.

Governments also play a central role through regulations, public investment, infrastructure development, research funding, energy policies, and international cooperation. The United Nations Environment Programme (UNEP, 2025) continues to assess the gap between countries' emission commitments and the reductions required to keep global warming consistent with internationally agreed temperature goals. Closing this emissions gap requires stronger

implementation rather than relying only on distant targets. Businesses also have responsibilities because corporate investment decisions influence energy systems, supply chains, transportation networks, manufacturing technologies, and patterns of resource consumption.

Adaptation is equally important because a certain degree of additional climate change is already unavoidable due to past and continuing emissions. Adaptation measures include heat-action plans, improved flood protection, stronger building standards, drought-resistant crops, efficient irrigation systems, wildfire preparedness, climate-resilient healthcare facilities, early-warning systems, urban green spaces, and improved disaster-management capabilities. Adaptation planning should prioritize populations facing the greatest risks and having the fewest resources to respond.

Individual behavior can contribute through energy conservation, lower-emission transportation, reduction of food and material waste, and sustainable consumption. Nevertheless, global warming cannot be solved solely through individual lifestyle choices. Large-scale transformation of energy, transportation, industry, agriculture, land use, and infrastructure is necessary.

Governments, businesses, communities, researchers, and individuals therefore have complementary responsibilities.

Ultimately, global warming represents one of the most significant long-term challenges facing humanity. The scientific evidence demonstrates clearly that human-generated greenhouse gas emissions are the dominant cause of recent warming, while observed changes in temperature, oceans, ice, weather extremes, ecosystems, and sea levels show that its consequences are already

occurring. Continued warming will increase risks to health, biodiversity, food and water security, economic development, infrastructure, and vulnerable communities. However, the severity of future impacts is not predetermined. Rapid reductions in greenhouse gas emissions, combined with effective adaptation, technological innovation, ecosystem protection, and international cooperation, can substantially reduce future damages. Global warming therefore demands immediate and sustained action based on scientific evidence, social responsibility, and recognition that decisions made today will influence environmental and human conditions for generations to come.

Global Warming: Causes, Consequences, and Global Responses

Global warming refers to the long-term increase in Earth's average surface temperature, primarily resulting from human activities that increase concentrations of heat-trapping greenhouse gases in the atmosphere. Although Earth's climate has always experienced natural variations, the warming observed since the Industrial Revolution differs substantially in its speed, scale, and principal causes. Scientific observations from surface monitoring stations, satellites, ocean measurements, glaciers, ice cores, and other sources provide strong evidence that the planet is warming and that human activities are the dominant cause. The Intergovernmental Panel on Climate Change (IPCC, 2023) concluded that human activities, principally greenhouse gas emissions, have unequivocally caused global warming. Average global surface temperature during 2011–2020 was approximately 1.1°C above the 1850–1900 average. This seemingly modest change has already produced significant effects on weather systems, ecosystems, human health, agriculture, water availability, and economies. Global

warming is therefore not simply an environmental concern; it is also a social, economic, developmental, and public-health challenge requiring coordinated action at local, national, and international levels.

Causes and Scientific Evidence of Global Warming

The principal cause of modern global warming is the enhanced greenhouse effect created by increasing concentrations of carbon dioxide, methane, nitrous oxide, and other greenhouse gases. The natural greenhouse effect is essential for life because gases in the atmosphere retain part of the heat that would otherwise escape into space. Human activity, however, has strengthened this process by releasing additional greenhouse gases. Fossil-fuel combustion for electricity, transportation, heating, industrial activity, and manufacturing remains a major source of carbon dioxide emissions. Agriculture, livestock production, waste management, fossil-fuel extraction, and other activities contribute substantial methane emissions, while agricultural fertilizers and certain industrial processes release nitrous oxide. Deforestation further contributes to the problem because removing forests releases stored carbon and reduces the ability of ecosystems to absorb carbon dioxide from the atmosphere.

The rapid expansion of industrial production and energy consumption since the nineteenth century has therefore altered the composition of the atmosphere. NASA explains that the current warming trend is clearly linked to human activities since the mid-1800s and is occurring at a rate not observed over many recent millennia (National Aeronautics and Space Administration [NASA], n.d.-a). Atmospheric carbon dioxide concentrations can be reconstructed through ice

cores and compared with modern direct measurements, providing scientists with a long-term record of environmental changes. These records demonstrate that recent increases in atmospheric carbon dioxide are substantially different from the natural fluctuations associated with previous climate cycles.

Natural factors such as volcanic eruptions, changes in solar activity, and variations in Earth's orbit can influence the climate. However, these factors cannot explain the magnitude and pattern of warming observed during the modern industrial era. The IPCC (2023) estimated that the best estimate of human-caused warming between 1850–1900 and 2010–2019 was approximately 1.07°C. In comparison, natural solar and volcanic drivers produced only very small changes in global temperature over the same broad period. The evidence therefore contradicts claims that contemporary warming can primarily be explained by natural climate cycles.

Multiple independent observations confirm the warming trend. Global surface-temperature datasets developed by different scientific organizations use somewhat different methodologies but consistently demonstrate substantial long-term warming. NASA reports that recent decades have become progressively warmer and that the temperature increase since 1970 has occurred faster than during any other 50-year period in at least the last 2,000 years (NASA, n.d.-b). Scientists also examine indirect indicators such as tree rings, coral formations, ice cores, sediments, and glacier records. Agreement among these independent sources significantly strengthens confidence in the scientific conclusion that the planet is warming.

Changes in the oceans and cryosphere provide additional evidence. Oceans absorb most of the additional heat accumulated in the climate system. According to the IPCC (2023), ocean warming accounted for approximately 91% of the heating within the climate system, while smaller proportions were associated with warming land, melting ice, and atmospheric warming. Glaciers and ice sheets have lost mass, Arctic sea ice has declined, and global sea levels have risen as a consequence of both melting land ice and the thermal expansion of seawater. Human influence has also contributed to changes in extreme weather conditions. Hot extremes and heat waves have become more frequent and intense across most land regions since the 1950s, while marine heat waves have approximately doubled in frequency since the 1980s (IPCC, 2023).

Global warming is consequently better understood as a disruption of the entire climate system rather than simply an increase in average temperature. Changes in atmospheric and ocean temperatures affect precipitation patterns, drought conditions, storms, ecosystems, ice, and sea levels. Because these components interact, warming can produce complex and cascading consequences. The scientific evidence gathered through independent measurements and decades of research demonstrates that limiting further warming requires substantial reductions in human-generated greenhouse gas emissions.

Environmental, Social, and Economic Impacts and Global Responses

The effects of global warming are already visible throughout natural and human systems. One of the clearest consequences is the increasing frequency and severity of extreme heat. Higher average temperatures make extreme heat events more probable and can increase their intensity

and duration. Heat waves create serious risks for older adults, children, outdoor workers, and people with existing illnesses. Extreme heat can cause dehydration, heat exhaustion, heat stroke, cardiovascular stress, and death. The World Health Organization (WHO, 2023) identifies climate change as a major threat to human health because it affects clean air, safe water, food supplies, shelter, livelihoods, and healthcare systems. Climate change is also associated with health risks arising from floods, storms, wildfires, droughts, infectious diseases, and mental-health pressures.

Agriculture and food security are particularly vulnerable because crop productivity depends on temperature, water availability, soil quality, predictable seasonal conditions, and protection from extreme weather. Excessive heat can reduce crop yields, while drought can create water shortages and damage livestock production. Conversely, intense rainfall and flooding can destroy crops, erode soil, damage transportation networks, and disrupt food distribution. Changes in climatic conditions may also allow agricultural pests and plant diseases to expand into new areas. These effects can increase food prices and intensify malnutrition, especially in low-income regions where households already spend a significant proportion of their income on food.

Water resources are similarly affected. Changing rainfall patterns may increase drought risks in some regions while increasing flooding in others. Mountain glaciers, which provide seasonal freshwater to many communities, are retreating as temperatures rise. Coastal freshwater supplies may also face saltwater intrusion as sea levels increase. Water scarcity can affect agriculture, electricity generation, sanitation, industry, and household consumption, potentially increasing competition over already limited resources.

Ecosystems are being transformed as species respond to changing temperatures, precipitation patterns, wildfire conditions, and habitat availability. Some species can migrate toward cooler regions or higher elevations, whereas others cannot adapt or move rapidly enough. Changes in species distributions can alter food webs and ecological relationships. Abrahms et al. (2023) found that climatic changes can also intensify human-wildlife conflicts by altering animal behavior, resource availability, and patterns of interaction between wildlife and human populations. These ecological disruptions matter because healthy ecosystems provide important services, including pollination, water regulation, soil protection, carbon storage, fisheries, and cultural benefits.

Coastal regions face another major threat from rising sea levels. Sea-level rise increases the likelihood of coastal flooding, erosion, saltwater intrusion, and damage from storm surges. Small island states, low-lying coastal communities, and densely populated river deltas face particularly serious risks. Global warming may therefore increase displacement and migration when communities lose homes, agricultural land, freshwater supplies, or livelihoods. Importantly, climate impacts are not distributed equally. The IPCC (2023) emphasizes that communities that have historically contributed the least to climate change are often disproportionately affected. Poverty, inadequate infrastructure, limited healthcare access, weak institutions, and geographical exposure can make communities significantly more vulnerable. Climate change therefore raises important questions about environmental justice and international responsibility.

Addressing global warming requires both mitigation and adaptation. Mitigation focuses on reducing the causes of climate change, whereas adaptation involves preparing societies and

ecosystems for impacts that can no longer be completely avoided. Effective mitigation requires reducing dependence on fossil fuels, expanding renewable and low-carbon energy, improving energy efficiency, electrifying appropriate forms of transportation and heating, reducing methane emissions, protecting forests, restoring degraded ecosystems, and improving agricultural and industrial practices. Technological innovation can support this transition through improved battery storage, electricity grids, renewable-energy systems, energy-efficient buildings, low-carbon industrial processes, and carbon-management technologies.

Governments also play a central role through regulations, public investment, infrastructure development, research funding, energy policies, and international cooperation. The United Nations Environment Programme (UNEP, 2025) continues to assess the gap between countries' emission commitments and the reductions required to keep global warming consistent with internationally agreed temperature goals. Closing this emissions gap requires stronger implementation rather than relying only on distant targets. Businesses also have responsibilities because corporate investment decisions influence energy systems, supply chains, transportation networks, manufacturing technologies, and patterns of resource consumption.

Adaptation is equally important because a certain degree of additional climate change is already unavoidable due to past and continuing emissions. Adaptation measures include heat-action plans, improved flood protection, stronger building standards, drought-resistant crops, efficient irrigation systems, wildfire preparedness, climate-resilient healthcare facilities, early-warning systems, urban green spaces, and improved disaster-management capabilities. Adaptation

planning should prioritize populations facing the greatest risks and having the fewest resources to respond.

Individual behavior can contribute through energy conservation, lower-emission transportation, reduction of food and material waste, and sustainable consumption. Nevertheless, global warming cannot be solved solely through individual lifestyle choices. Large-scale transformation of energy, transportation, industry, agriculture, land use, and infrastructure is necessary. Governments, businesses, communities, researchers, and individuals therefore have complementary responsibilities.

Ultimately, global warming represents one of the most significant long-term challenges facing humanity. The scientific evidence demonstrates clearly that human-generated greenhouse gas emissions are the dominant cause of recent warming, while observed changes in temperature, oceans, ice, weather extremes, ecosystems, and sea levels show that its consequences are already occurring. Continued warming will increase risks to health, biodiversity, food and water security, economic development, infrastructure, and vulnerable communities. However, the severity of future impacts is not predetermined. Rapid reductions in greenhouse gas emissions, combined with effective adaptation, technological innovation, ecosystem protection, and international cooperation, can substantially reduce future damages. Global warming therefore demands immediate and sustained action based on scientific evidence, social responsibility, and recognition that decisions made today will influence environmental and human conditions for generations to come.

Global Warming: Causes, Consequences, and Global Responses

Global warming refers to the long-term increase in Earth's average surface temperature, primarily resulting from human activities that increase concentrations of heat-trapping greenhouse gases in the atmosphere. Although Earth's climate has always experienced natural variations, the warming observed since the Industrial Revolution differs substantially in its speed, scale, and principal causes. Scientific observations from surface monitoring stations, satellites, ocean measurements, glaciers, ice cores, and other sources provide strong evidence that the planet is warming and that human activities are the dominant cause. The Intergovernmental Panel on Climate Change (IPCC, 2023) concluded that human activities, principally greenhouse gas emissions, have unequivocally caused global warming. Average global surface temperature during 2011–2020 was approximately 1.1°C above the 1850–1900 average. This seemingly modest change has already produced significant effects on weather systems, ecosystems, human health, agriculture, water availability, and economies. Global warming is therefore not simply an environmental concern; it is also a social, economic, developmental, and public-health challenge requiring coordinated action at local, national, and international levels.

Causes and Scientific Evidence of Global Warming

The principal cause of modern global warming is the enhanced greenhouse effect created by increasing concentrations of carbon dioxide, methane, nitrous oxide, and other greenhouse gases. The natural greenhouse effect is essential for life because gases in the atmosphere retain part of

the heat that would otherwise escape into space. Human activity, however, has strengthened this process by releasing additional greenhouse gases. Fossil-fuel combustion for electricity, transportation, heating, industrial activity, and manufacturing remains a major source of carbon dioxide emissions. Agriculture, livestock production, waste management, fossil-fuel extraction, and other activities contribute substantial methane emissions, while agricultural fertilizers and certain industrial processes release nitrous oxide. Deforestation further contributes to the problem because removing forests releases stored carbon and reduces the ability of ecosystems to absorb carbon dioxide from the atmosphere.

The rapid expansion of industrial production and energy consumption since the nineteenth century has therefore altered the composition of the atmosphere. NASA explains that the current warming trend is clearly